

Ultimate Nutrition

TAURINE

Taurine is a versatile amino acid that contains a sulfur group rather than a carboxyl group. Taurine is a multifaceted amino. Like glutamine, it can be described as a jack-of-all-trades. Scientists identified a role for taurine in nutrition only about 23 years ago. Taurine is involved in insulin-like activity as well as detoxification, protection of cell membranes, regulation of blood pressure and the maintenance of structural proteins in muscle.

Some scientists also consider taurine a conditionally essential amino acid. Although we normally can make a certain amount of taurine, deficiency can manifest itself in functional impairment of certain tissues; taurine may act as a growth modulator. Nevertheless, the role of taurine in human nutrition isn't completely understood. But with its multiple functions, it comes as no surprise that perhaps this amino can benefit bodybuilders.

From a bodybuilding standpoint, nothing is more important than maintaining an anabolic state. Bodybuilders eat 5-6 times per day and maintain a high protein and carbohydrate intake while attempting to limit fat intake. They train intensely and recover sufficiently in the hope of increasing muscle protein.

More than 50 years ago, scientists discovered that taurine had an effect similar to that of insulin on carbohydrate metabolism - it acted as a strong hypoglycemic agent. In 1964 this amino was found to increase glucose utilization in the diaphragm muscles of rats. Insulin produces a drop in blood-glucose levels and transports glucose into cells. Insulin promotes the influx of amino acids and glucose into muscle tissue.

Taurine mimics the action of insulin both in vivo (in living organisms) and in vitro (test tubes). In one experiment, it stimulated glucose uptake into cells at 50% the capacity of insulin. Insulin and taurine also similarly increase the uptake of the amino acid proline into rat diaphragms. Interestingly, the combination of insulin and taurine was no better than either insulin or taurine by itself. Furthermore, like insulin, taurine can stimulate the formation of glycogen in cardiac muscle, certainly an important aspect of training.

A high concentration of taurine exists in skeletal muscles. Since most taurine in skeletal muscle is present as a free amino acid, a physiological role probably exists. Taurine has been shown to participate in the excitation-contraction coupling mechanism in skeletal muscle, which means that it affects the transmission of an electrical signal into muscle fibers. This has obvious importance in ensuring optimal muscle performance.

Since taurine is a multifaceted amino acid, you shouldn't be too surprised to learn that it affects blood pressure. Researchers used rats bred to have high blood pressure to study the effects of differing concentration of taurine on food intake, bodyweight and blood pressure. Taurine was administered in the animals' drinking water either as a 1% or 3%

solution over a period lasting eight weeks. The amino had no effect on bodyweight and food intake, but blood pressure decreased by the fourth week of treatment and remained lower than the untreated group throughout the treatment period. The 3% taurine solution had a greater effect on blood pressure than the 1% solution. Taurine-treated animal had significantly higher tissue level in kidney and skeletal muscle, yet level in certain parts of the brain didn't change.

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